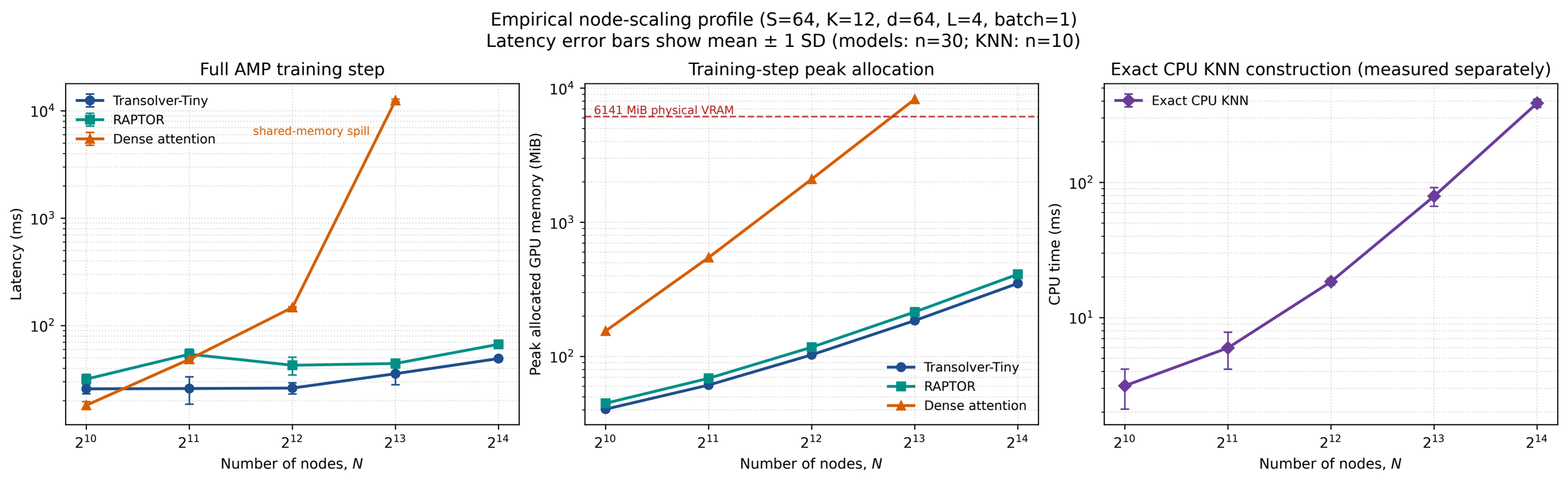

Empirical node-scaling profile (S=64, K=12, d=64, L=4, batch=1)
Latency error bars show mean ± 1 SD (models: n=30; KNN: n=10)
Full AMP training step
Transolver-Tiny
RAPTOR
Dense attention
shared-memory spill
Latency (ms)
Number of nodes, N
Training-step peak allocation
6141 MiB physical VRAM
Peak allocated GPU memory (MiB)
Transolver-Tiny
RAPTOR
Dense attention
Number of nodes, N
Exact CPU KNN construction (measured separately)
Exact CPU KNN
CPU time (ms)
Number of nodes, N